

Data Processing on Modern Hardware

Assignment 5 – Synchronization

Handout: 4th June 2025
Due: 11th June 2025 by 14:00

Introduction

The goal of this exercise is to compare different synchronization approaches introduced in this week's lecture. We analyze pessimistic and optimistic locking, exclusive and shared locks and vary the granularity from coarse- to fine-grained locking. As workload we use a list-based set provided in the code skeleton.

Part 1 - Implementation

Your task is synchronizing the given list-based set using the following approaches:

1. Coarse-Grained Locking
2. Coarse-Grained Locking with read/write lock
3. Lock Coupling
4. Lock Coupling with read/write locks
5. Optimistic
6. Optimistic Lock Coupling

For the purpose of this assignment, you are also allowed to ignore memory reclamation (do not call delete).

This week's lecture already contains all needed information to implement the different approaches. If you still need additional information you can have a look into the book "The Art of Multiprocessor Programming"¹.

Part 2 - Analysis and Evaluation

Apply the different synchronization techniques for the list-based set to analyze and compare them:

- How expensive is locking for the different approaches?
- How do the approaches perform in regard to #clock-cycles, #instructions and IPC?
- Which approach provides the best performance?

Vary the number of threads from low- to high contention, e.g., from 1-20 threads on the chair's SkylakeX machine. Further, investigate the effect of the domain parameter. Provide graphical plots for your results and normalize the results to make them comparable, e.g., using parallel speed-up as metric (lecture 7, slide 6).

¹<https://dl.acm.org/doi/pdf/10.5555/2385452>

Submission guidelines

This homework has a duration of one week. Fork the repository and commit your changes in the git to hand in your homework. You are expected to implement the aforementioned synchronization methods and write a report summarizing your findings from part 2.

The programming language of this homework is C++ or C. We provide you a code skeleton, add functions needed for the implementation. For performance measuring of the experiments, you can either use the provided `perfEvent.hpp` and the commented code blocks or use the tools you applied in the previous assignments.